

What the system protects and what it assumes

The contracts control calculation and result release, but do not prove biological origin or clinical validity

What the contracts enforce

result rules fixed by model provider
valid calculation order
numeric score only to intended requester
no public numeric-score release
one result per calculation + query limits

Possible attacker actions

read public blockchain data;
operate a blockchain node; submit transactions;
make repeated requests with chosen inputs

Not established by the system

whether genotypes came from the sample;
model validity and calibration;
performance across ancestry groups;
formal privacy guarantee

Assumptions and services the system depends on

security of TFHE encryption; deployed contract code; blockchain consensus;
fhEVM services that calculate encrypted values, submit transactions, and manage keys